GCSE BIOLOGY

Higher Tier – Paper 1 style

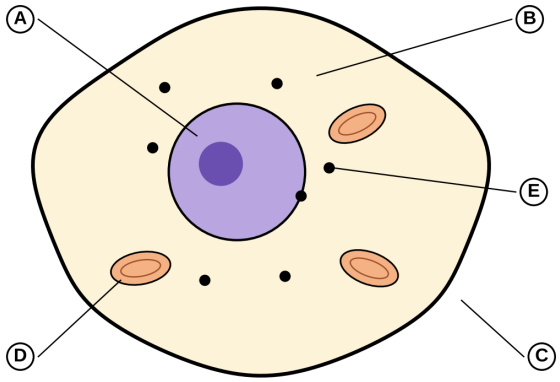
Topic 4.1: Cell biology

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 185 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **176**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 176

0 1 The diagram shows an animal cell. The letters **A** to **E** show different parts of the cell.



0 1 . 1 Name the parts labelled **A** to **E**. **[5 marks]**

0 1 . 2 Give the function of part **A**. **[1 mark]**

0 1 . 3 Give the function of part **D**. **[1 mark]**

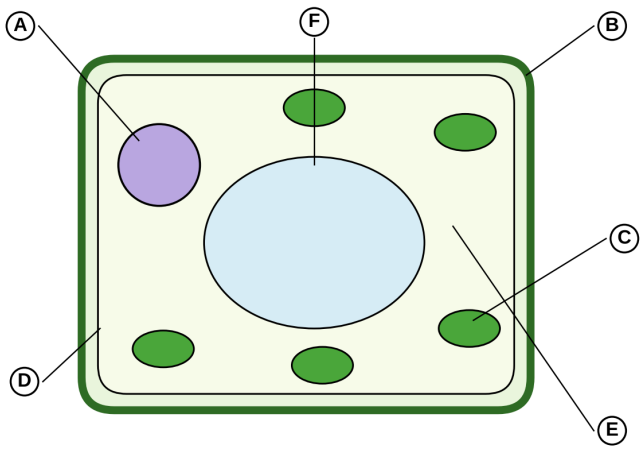
0 1 . 4 Give the function of part **E**. **[1 mark]**

0 1 . 5 Part **C** controls the movement of substances into and out of the cell. Explain why the cell needs this control. **[2 marks]**

0 1 . 6 Muscle cells have many more mitochondria than skin cells. Explain why. **[2 marks]**

0 1 . 7 Animal cells are eukaryotic. What is meant by a **eukaryotic** cell? **[1 mark]**

0 2 The diagram shows a plant cell from a leaf. The letters **A** to **F** show different parts of the cell.



0 2 . 1 Draw one line from each part to its function. **[4 marks]**

Part	Function
cell wall	Contains cell sap
chloroplast	Controls the activities of the cell
permanent vacuole	Absorbs light for photosynthesis
nucleus	Strengthens the cell

0 2 . 2 Name the parts labelled **A** to **F**. **[6 marks]**

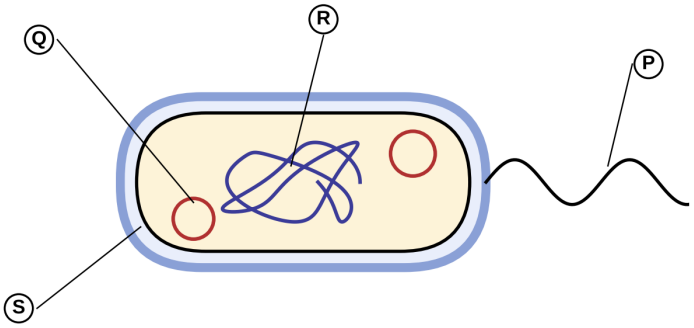
0 2 . 3 Name the substance that cell walls are made of. **[1 mark]**

0 2 . 4 Name **two** parts found in a plant cell but not in an animal cell. **[2 marks]**

0 2 . 5 Root cells do not contain chloroplasts. Explain why. **[2 marks]**

0 2 . 6 Explain why a plant cell does not burst when it takes in a lot of water. **[2 marks]**

0 3 The diagram shows a bacterial cell. The letters **P** to **S** show different parts of the cell.



0 3 . 1 Name the parts labelled **P**, **Q**, **R** and **S**. **[4 marks]**

0 3 . 2 Bacteria are prokaryotic cells. Give **two** differences between a bacterial cell and an animal cell. **[2 marks]**

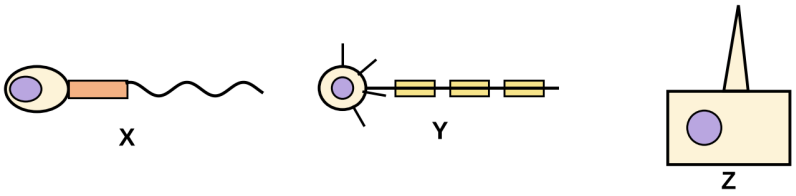
0 3 . 3 A bacterial cell is 2 μm long. Write this length in metres using standard form. **[2 marks]**

Answer = _____ m

0 3 . 4 An animal cell is 20 μm wide. How many times wider is the animal cell than the bacterial cell? **[1 mark]**

0 3 . 5 State the function of the **flagellum** in a bacterial cell. **[1 mark]**

0 4 The diagram shows three specialised cells: **X** (sperm cell), **Y** (nerve cell) and **Z** (root hair cell).



0 4 . 1 What is meant by **differentiation**? **[1 mark]**

0 4 . 2 Describe **two** ways cell **X** is adapted for its function. **[2 marks]**

0 4 . 3 Explain how cell **Y** is adapted to carry electrical impulses. **[2 marks]**

0 4 . 4 Explain how cell **Z** is adapted to take in water and mineral ions from the soil. **[3 marks]**

0 4 . 5 Most animal cells differentiate early in life. Describe how this is different in plant cells. **[2 marks]**

0 5 A student looks at a cell using a light microscope and draws what she sees.

0 5 . 1 The student's drawing of the cell is 36 mm long. The real cell is 12 µm long. Calculate the magnification of the drawing. [1 mm = 1000 µm] **[3 marks]**

Answer = _____

0 5 . 2 A different image of a cell is 45 mm long and the magnification is × 1500. Calculate the real length of the cell. Give your answer in µm. **[3 marks]**

Answer = _____ µm

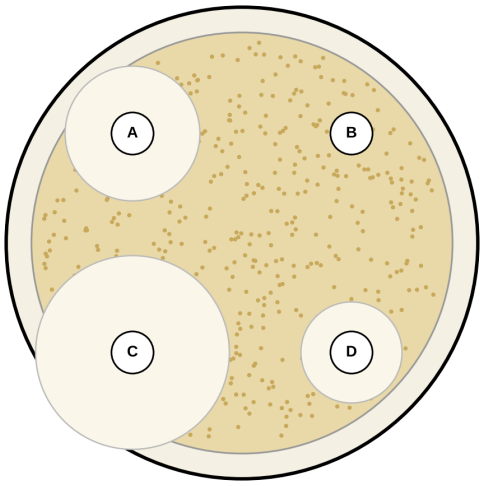
0 5 . 3 An electron microscope can show structures such as ribosomes. A light microscope cannot. Explain why. **[2 marks]**

0 5 . 4 Give **two** advantages of using an electron microscope instead of a light microscope. **[2 marks]**

0 5 . 5 Suggest **one** advantage of using a light microscope instead of an electron microscope. **[1 mark]**

0 5 . 6 Describe how the student should prepare a slide of onion cells to view under a light microscope. **[4 marks]**

0 6 A student tests four antibiotics, **A**, **B**, **C** and **D**, on bacteria. She spreads bacteria on agar jelly in a Petri dish and adds a paper disc soaked in each antibiotic. The dish is incubated. The diagram shows the dish. The table shows the diameter of the clear zone around each disc.



0 6 . 1 The student sterilised the Petri dish and agar before use. Explain why. **[2 marks]**

Antibiotic	**A**	**B**	**C**	**D**
Diameter of clear zone in mm	12	0	20	8

0 6 . 2 Describe **two** other precautions the student should take to avoid contamination. **[2 marks]**

0 6 . 3 The dish is incubated at 25 °C. Explain why a higher temperature, such as 37 °C, is not used in school. **[2 marks]**

0 6 . 4 Which antibiotic is the most effective at killing the bacteria? Give a reason. **[2 marks]**

0 6 . 5 Antibiotic **B** has no clear zone. What does this tell you about the bacteria? **[1 mark]**

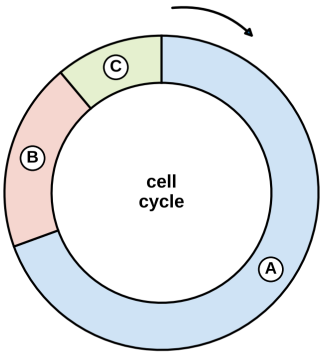
0 6 . 6 Calculate the area of the clear zone around disc **A**. **[3 marks]**
Area = πr^2 ($\pi = 3.14$)

Answer = _____ mm²

0 6 . 7 A single bacterium divides by binary fission every 20 minutes. Calculate the number of bacteria after 4 hours. Give your answer in standard form to 2 significant figures. **[3 marks]**

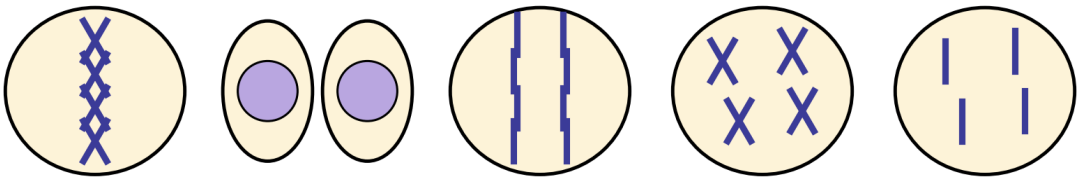
Answer = _____

0 7 The diagram shows the cell cycle. Diagrams **A** to **E** show what happens to the chromosomes of an animal cell with four chromosomes during one cell cycle.



0 7 . 1 What happens to the DNA in the cell before mitosis starts? **[2 marks]**

0 7 . 2 Put diagrams **A** to **E** in the correct order. Start with E. **[2 marks]**



0 7 . 3 Describe what happens during mitosis. **[3 marks]**

0 7 . 4 A human body cell has 46 chromosomes. How many chromosomes will be in each new cell made by mitosis? **[1 mark]**

0 7 . 5 Give **two** reasons why organisms need mitosis. **[2 marks]**

0 7 . 6 Cancer is caused by changes in cells that lead to uncontrolled growth and division. Explain why this results in a tumour. **[2 marks]**

0 8 This question is about stem cells.

0 8 . 1 What is a **stem cell**? **[2 marks]**

0 8 . 2 Name the **two** sources of stem cells in humans. **[2 marks]**

0 8 . 3 Name the part of a plant where stem cells are found. **[1 mark]**

0 8 . 4 Give **one** use of plant stem cells. **[1 mark]**

0 8 . 5 Stem cells from embryos could be used to treat people with diabetes. Explain how. **[2 marks]**

0 8 . 6 Give **two** risks of using stem cells in medicine. **[2 marks]**

0 8 . 7 Some people object to the use of embryonic stem cells. Suggest **one** reason. **[1 mark]**

0 8 . 8 What is **therapeutic cloning**? Explain why it is useful. **[3 marks]**

0 9 Diagram 1 shows particles of a substance on either side of a cell membrane.

0 9 . 1 What is meant by **diffusion**? **[2 marks]**

0 9 . 2 In which direction will there be a net movement of particles in Diagram 1? Give a reason. **[2 marks]**

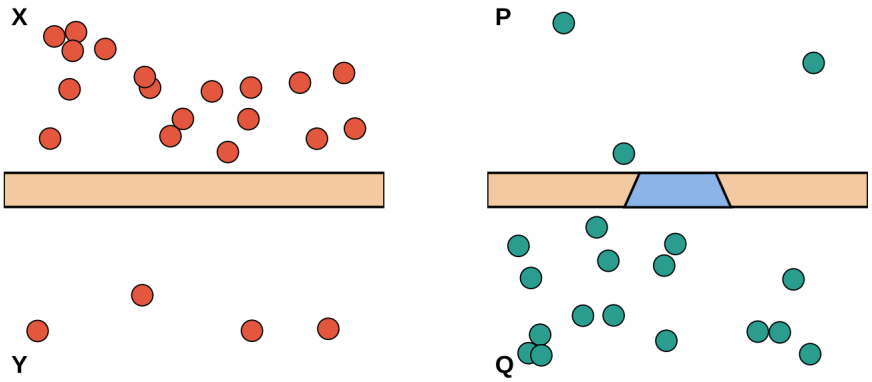


Diagram 1

Diagram 2

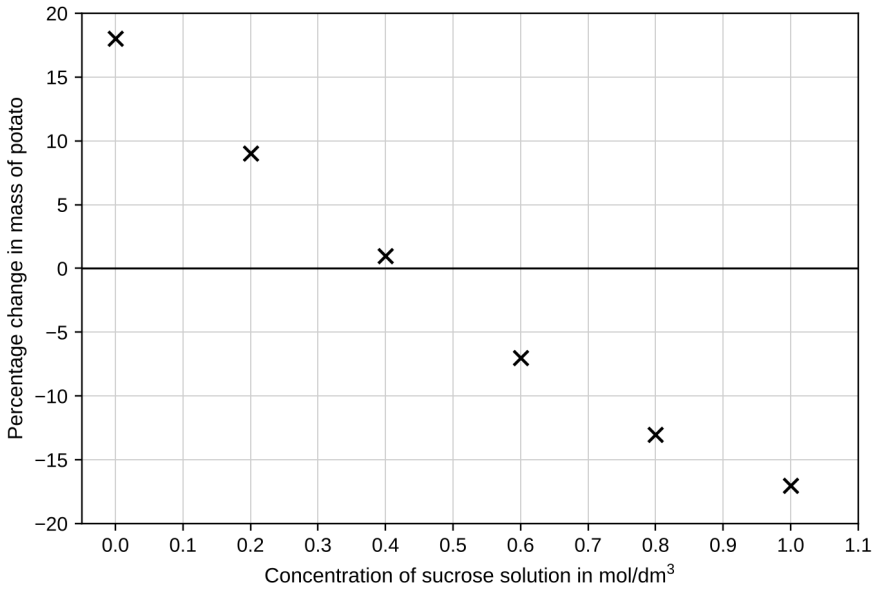
0 9 . 3 Give **three** factors that affect the rate of diffusion. **[3 marks]**

0 9 . 4 Explain why the rate of diffusion increases when the temperature increases. **[2 marks]**

0 9 . 5 Name **one** substance that moves into cells by diffusion and **one** that moves out. **[2 marks]**

0 9 . 6 Explain why the cell membrane is described as **partially permeable**. **[2 marks]**

1 0 A student investigates osmosis. She cuts potato into identical cylinders, finds the mass of each, and puts one in each of six sucrose solutions. After 24 hours she finds the mass again and calculates the percentage change in mass. The graph shows her results.



1 0 . 1 Define **osmosis**. **[2 marks]**

1 0 . 2 A potato cylinder has an initial mass of 5.0 g and a final mass of 5.9 g. Calculate the percentage change in mass. **[3 marks]**

Answer = _____ %

1 0 . 3 Draw a line of best fit through the points. **[1 mark]**

10.4 Use your line to find the concentration of the sucrose solution that has the same concentration as the inside of the potato cells. **[2 marks]**

10.5 Explain why the potato cylinder in 1.0 mol/dm^3 sucrose solution decreases in mass. **[3 marks]**

10.6 Give **two** variables the student should control. **[2 marks]**

10.7 The student blots the potato cylinders dry before finding the final mass. Explain why. **[1 mark]**

11 Diagram 2 (shown earlier) shows a cell membrane with a carrier protein. The concentration of the substance is higher at **Q** than at **P**.

11.1 What is meant by **active transport**? **[2 marks]**

11.2 Diagram 2 shows particles being moved. In which direction will particles move: P to Q or Q to P? Give a reason. **[2 marks]**

1 1 . 3 Root hair cells take in mineral ions from the soil by active transport. Explain why root hair cells need many mitochondria. **[2 marks]**

1 1 . 4 Explain why mineral ions cannot enter root hair cells by diffusion. **[1 mark]**

1 1 . 5 Give **one** other example of active transport in the human body. **[1 mark]**

1 1 . 6 Complete the table to show **two** differences between diffusion and active transport. **[2 marks]**

	Diffusion	**Active transport**
Energy from respiration needed?		
Direction relative to the concentration gradient		

1 2 Cells and organisms need to exchange substances with their surroundings.

1 2 . 1 Calculate the surface area to volume ratio of a cube with sides of 2 cm. Give your answer in the form **x : 1**. **[3 marks]**

Answer = _____ : 1

1 2 . 2 Calculate the surface area to volume ratio of a cube with sides of 6 cm. **[3 marks]**

Answer = _____ : 1

1 2 . 3 What happens to the surface area to volume ratio as an organism gets larger? **[1 mark]**

1 2 . 4 A single-celled organism can exchange substances across its cell membrane. Explain why a large multicellular organism needs special exchange surfaces. **[3 marks]**

1 2 . 5 Name **one** exchange surface in humans and give **two** features that make it efficient. **[3 marks]**

1 2 . 6 Suggest how the structure of a fish gill makes it an efficient exchange surface. **[2 marks]**

1 3 Stretch questions (grade 7–9).

1 3 . 1 *Compare the structure of a bacterial cell with the structure of an animal cell. Explain why bacteria are described as prokaryotes. **[6 marks]**

1 3 . 2 *Describe how you would investigate the effect of different concentrations of sugar solution on the mass of potato cylinders. Explain how you would process your results to find the concentration of the potato cells. **[6 marks]**

1 3 . 3 *Evaluate the use of embryonic stem cells to treat disease. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.1: Cell biology

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	A: nucleus ; B: cytoplasm ; C: (cell) membrane ; D: mitochondrion / mitochondria ; E: ribosome ;	One mark for each correct name.	5	AO1 4.1.1.2
01.2	contains genetic material / DNA / chromosomes ; controls the activities of the cell	Any one.	1	AO1 4.1.1.2
01.3	where aerobic respiration takes place / releases energy		1	AO1 4.1.1.2
01.4	where protein synthesis happens / proteins are made		1	AO1 4.1.1.2
01.5	to let in substances the cell needs (e.g. oxygen, glucose) ; to keep out harmful substances / keep in useful substances ;		2	AO2 4.1.1.2
01.6	muscle cells need a lot of energy (for contraction) ; mitochondria release energy by respiration ;		2	AO2 4.1.1.2
01.7	a cell with a nucleus (that contains genetic material)		1	AO1 4.1.1.1
02.1	cell wall — strengthens the cell / supports the cell ; chloroplast — absorbs light for photosynthesis ; permanent vacuole — contains cell sap ; nucleus — controls the cell's activities / contains DNA ;	One mark for each correct link.	4	AO1 4.1.1.2
02.2	A: nucleus ; B: cell wall ; C: chloroplast ; D: (cell) membrane ; E: cytoplasm ; F: (permanent) vacuole ;	One mark for each correct name.	6	AO1 4.1.1.2
02.3	cellulose		1	AO1 4.1.1.2
02.4	(permanent) vacuole ; chloroplast / cell wall ;	Any two from: cell wall, chloroplasts, permanent vacuole.	2	AO1 4.1.1.2
02.5	no light reaches the roots / they are underground ; so photosynthesis cannot happen there ;		2	AO2 4.1.1.2
02.6	the cell wall is strong / rigid ; it stops the cell expanding too far ;		2	AO2 4.1.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	P: flagellum ; Q: plasmid ; R: (circular) chromosome / DNA loop ; S: cell wall ;	Allow 'flagella' for P.	4	AO1 4.1.1.2
03.2	bacterial cell has no nucleus / DNA is a single loop ; bacterial cell has a cell wall / plasmids ;	Allow: much smaller; no mitochondria.	2	AO1 4.1.1.1
03.3	2×10^{-6} ; m (1 mark for correct power, 1 for the correct value) ;		2	AO2 4.1.1.1
03.4	10 times (20 ÷ 2)		1	AO2 4.1.1.1
03.5	helps the bacterium to move / swim		1	AO1 4.1.1.2
04.1	the process by which a cell changes to become specialised for its job		1	AO1 4.1.1.3
04.2	has a tail (flagellum) to swim ; has many mitochondria to provide energy ; has enzymes in the head to digest the egg membrane / streamlined head ;	Any two.	2	AO2 4.1.1.3
04.3	it is long ; it has branched endings (dendrites) to make connections with other nerve cells ; it has an insulating myelin sheath to speed up impulses ;	Any two.	2	AO2 4.1.1.3
04.4	long hair-like projection increases surface area ; so more water is absorbed (by osmosis) ; many mitochondria for active transport of mineral ions ;		3	AO2 4.1.1.3
04.5	many types of plant cell can differentiate throughout life ; this allows them to form new roots, stems or leaves / clones ;		2	AO1 4.1.1.3
05.1	36 mm = 36 000 µm ; 36 000 ÷ 12 ; × 3000 ;	magnification = size of image ÷ size of real object. Correct answer with no working scores 3.	3	AO2 4.1.1.5
05.2	real size = image size ÷ magnification ; 45 ÷ 1500 = 0.03 mm ; 30 µm ;		3	AO2 4.1.1.5
05.3	an electron microscope has a higher resolution / resolving power ; so it can show smaller (more detailed) structures ;		2	AO2 4.1.1.4
05.4	higher magnification ; higher resolution (shows more detail of sub-cellular structures) ;		2	AO1 4.1.1.4
05.5	can view living cells / cheaper / portable / easy to use / quick to prepare slides	Any one.	1	AO3 4.1.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	place a thin layer of onion epidermis on a slide ; add a drop of water / stain (iodine) ; lower a coverslip at an angle (to avoid air bubbles) ; place on the stage and focus using the lowest-power lens first ;		4	AO1 4.1.1.4
06.1	to kill (unwanted) microorganisms ; so that only the bacteria being tested grow (no contamination) ;		2	AO2 4.1.1.6
06.2	flame the inoculating loop (to sterilise it) ; tape the lid on the dish (not completely sealed) / keep the lid on as much as possible ;	Allow 'work near a Bunsen flame'; 'disinfect the bench'. Any two.	2	AO1 4.1.1.6
06.3	at 37 °C harmful human pathogens could grow ; 25 °C reduces the chance of this ;		2	AO2 4.1.1.6
06.4	C ; it has the largest clear zone ;		2	AO3 4.1.1.6
06.5	the bacteria are resistant to antibiotic B / it does not kill them		1	AO3 4.1.1.6
06.6	radius = 12 ÷ 2 = 6 mm ; 3.14×6^2 ; 113 mm^2 ;	Allow 113.04 or 113.1.	3	AO2 4.1.1.6
06.7	number of divisions = 12 ; $2^{12} = 4096$; 4.1×10^3 ;	Assume the number doubles at each division.	3	AO2 4.1.1.6
07.1	the DNA is copied / replicated ; so each chromosome has two identical copies / the number of sub-cellular structures increases ;		2	AO1 4.1.2.1
07.2	E, D, A, C, B ; (four in correct order for 1 mark) ;	E: single chromosomes; D: copies made; A: chromosomes line up; C: pulled apart; B: cell divides.	2	AO2 4.1.2.1
07.3	the chromosomes line up at the centre of the cell ; the copies are pulled apart to opposite ends of the cell ; the nucleus divides then the cytoplasm and membranes divide to form two identical cells ;		3	AO1 4.1.2.1
07.4	46		1	AO2 4.1.2.1
07.5	growth ; repair of damaged tissue / replacement of cells ; asexual reproduction ;	Any two.	2	AO1 4.1.2.1
07.6	cells divide by mitosis too quickly / without control ; forming a mass of abnormal cells ;		2	AO2 4.1.2.1
08.1	an undifferentiated cell ; that can divide / differentiate into other types of cell ;		2	AO1 4.1.2.2
08.2	embryos ; adult bone marrow ;	Allow 'umbilical cord blood' for adult sources.	2	AO1 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.3	meristem (tissue)		1	AO1 4.1.2.2
08.4	to produce clones of plants quickly / cheaply ; to grow rare species / disease-resistant crops ;	Any one.	1	AO1 4.1.2.2
08.5	stem cells can be made to differentiate into insulin-producing cells ; to replace the damaged cells in the pancreas ;		2	AO2 4.1.2.2
08.6	stem cells may transfer a virus to the patient ; stem cells may divide out of control / cause tumours ; patient's body may reject the cells ;	Any two.	2	AO1 4.1.2.2
08.7	the embryo is destroyed / a potential human life ; religious / ethical beliefs ;	Any one.	1	AO3 4.1.2.2
08.8	an embryo with the same genes as the patient is made ; stem cells from the embryo are not rejected ; because they have the same genes / are genetically identical to the patient ;	Triple only (Higher Tier detail).	3	AO2 4.1.2.2
09.1	the (net) movement of particles ; from a region of higher concentration to a region of lower concentration ;		2	AO1 4.1.3.1
09.2	from X to Y ; X has a higher concentration than Y / down a concentration gradient ;		2	AO2 4.1.3.1
09.3	difference in concentration (concentration gradient) ; temperature ; surface area of the membrane ;		3	AO1 4.1.3.1
09.4	particles have more (kinetic) energy / move faster ; so more collisions / move across the membrane more quickly ;		2	AO2 4.1.3.1
09.5	into: oxygen / glucose / amino acids ; out: carbon dioxide / urea ;		2	AO1 4.1.3.1
09.6	it lets some substances through ; but not others (e.g. large molecules) ;		2	AO1 4.1.3.1
10.1	the movement of water ; from a dilute solution to a concentrated solution through a partially permeable membrane ;		2	AO1 4.1.3.2
10.2	5.9 − 5.0 = 0.9 g ; 0.9 ÷ 5.0 × 100 ; +18 % ;	Allow 18 % increase.	3	AO2 4.1.3.2
10.3	a smooth line / straight line with a balance of points either side		1	AO2 WS 3.5
10.4	where the line crosses 0 % change ; 0.40 – 0.44 mol/dm^3 ;		2	AO3 4.1.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.5	the solution is more concentrated than the cell contents ; water moves out of the potato cells (by osmosis) ; from a dilute solution to a more concentrated solution ;		3	AO3 4.1.3.2
10.6	volume of sucrose solution ; size / surface area / mass of potato cylinders / time / temperature / same type of potato ;	Any two.	2	AO3 4.1.3.2
10.7	to remove water on the surface (so the mass is just the potato)		1	AO3 4.1.3.2
11.1	movement of substances from a more dilute solution to a more concentrated solution (against a concentration gradient) ; using energy from respiration ;		2	AO1 4.1.3.3
11.2	P to Q ; particles are moving against the concentration gradient / carrier proteins move them ;		2	AO3 4.1.3.3
11.3	active transport needs energy ; mitochondria release energy by respiration ;		2	AO2 4.1.3.3
11.4	the concentration of ions is higher inside the root hair cell than in the soil		1	AO3 4.1.3.3
11.5	glucose absorbed from the gut / kidney tubules into the blood (when concentration is low)		1	AO1 4.1.3.3
11.6	diffusion: does not need energy from respiration ; diffusion: moves down a concentration gradient ; active transport: needs energy ; moves against the gradient ;	Mark each row separately; both columns needed per row.	2	AO1 4.1.3.3
12.1	surface area = 6 × 2 × 2 = 24 cm^2 ; volume = 2 × 2 × 2 = 8 cm^3 ; 24 : 8 = 3 : 1 ;		3	AO2 4.1.3.4
12.2	surface area = 6 × 6 × 6 = 216 cm^2 ; volume = 216 cm^3 ; 1 : 1 ;		3	AO2 4.1.3.4
12.3	it decreases		1	AO2 4.1.3.4
12.4	large organisms have a small surface area to volume ratio ; diffusion distance is too long / not enough surface area to supply all cells ; so they need specialised exchange surfaces and a transport system ;		3	AO2 4.1.3.4
12.5	alveoli (lungs) / villi (small intestine) ; large surface area ; thin membrane (short diffusion distance) / good blood supply / ventilation ;		3	AO1 4.1.3.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.6	many gill filaments / lamellae give a large surface area ; thin surface and good blood supply keep the concentration gradient high ;		2	AO3 4.1.3.4
13.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • animal cells are eukaryotic: DNA is contained within a nucleus; bacteria have no nucleus • bacterial DNA is a single circular chromosome in the cytoplasm; extra DNA in plasmids • both have cell membrane, cytoplasm and ribosomes • bacteria have a cell wall (not made of cellulose); animal cells do not • animal cells have mitochondria and other membrane-bound organelles; bacteria do not • bacteria are much smaller (about 1–2 μm compared with 10–30 μm); prokaryote means 'before nucleus' - no nucleus		6	AO1 4.1.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • cut identical cylinders of potato using a cork borer; same length / surface area • measure the initial mass of each cylinder; put one in each of several concentrations of sugar solution (same volume) • leave for the same time (e.g. 24 hours) at the same temperature • remove the cylinders, blot dry and measure the final mass • calculate the percentage change in mass for each; plot the percentage change against concentration • the concentration where the line crosses 0% change is the concentration of the potato cells (no net osmosis)		6	AO3 4.1.3.2
13.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • stem cells can differentiate into many types of cell so could treat many conditions (diabetes, paralysis, blindness) • they could replace damaged cells and give a long-term cure for some diseases • therapeutic cloning produces genetically identical cells, so there is less risk of rejection • risks: tumours, viral transfer from the donor cells, possible rejection • ethical issues: embryos are destroyed; some people think this is the loss of a potential life • balanced judgement, e.g. potential benefits are large and adult / meristem stem cells have fewer ethical objections but are less versatile		6	AO3 4.1.2.2

Where marks are lost in Topic 4.1

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Name AND function:** mitochondria = respiration (releases energy), ribosomes = protein synthesis, nucleus = controls the cell/contains DNA, chloroplast = photosynthesis. Do not say 'makes energy' — energy is released.
- ☐ **Magnification = image ÷ real size.** Convert units first (1 mm = 1000 µm). Write the working out and units; check by estimating if the answer is sensible.
- ☐ **Standard form:** 2 µm = 2×10^{-6} m. Practise converting between mm, µm and m.
- ☐ **Aseptic technique:** sterilise the loop, dish and agar, keep the lid on, tape the lid (not fully sealed), incubate at 25 °C (not 37 °C) at school.
- ☐ **Clear zone area** = πr^2. Use the radius (half the diameter). Compare zones to rank antibiotics: larger clear zone = more effective.
- ☐ **Mitosis:** DNA copied → chromosomes line up → copies pulled apart → two identical cells. Link it to growth, repair and asexual reproduction.
- ☐ **Osmosis:** water moves from a dilute solution (high water concentration) to a concentrated solution (low water concentration). Always say 'water' moves, not the solute, and mention the partially permeable membrane.
- ☐ **Active transport:** needs energy from respiration and goes against the concentration gradient. Link 'many mitochondria' to it.
- ☐ **Surface area : volume:** cube = $6 \times \text{side}^2 : \text{side}^3$. As organisms get bigger, the ratio gets smaller, so exchange surfaces are needed.
- ☐ **Stem cells:** always give one benefit and one risk/ethical issue for 6-mark 'evaluate' questions, then a judgement.